# ECOCHRONICLES

*A Gamified, AI-Assisted Environmental Education Platform for Schools*

Built to run at zero infrastructure cost — powered entirely by in-browser AI and Firebase's free tier.

**17-Level Quiz** **In-Browser AI Tutor** **Live Leaderboard** **Teacher Analytics**

# The Problem

*Environmental education is losing to boredom — and budgets*



### Passive Learning

Textbook-only environmental lessons don't stick — students disengage fast, and knowledge rarely turns into behavior.



### Budget-Starved Schools

Most EdTech "AI" tools require paid API keys or cloud AI subscriptions schools simply can't afford.



### No Visibility for Teachers

Teachers have no easy way to see who's struggling, what topics need reteaching, or class-wide progress.

**The result:** students who can pass a quiz but can't tell you what to do with a dead battery.

# The Solution

*EcoChronicles turns environmental learning into a game — with a personal AI tutor, at zero cost*

17-level gamified environmental quiz — Beginner to Advanced

Shelly, an offline AI Eco-Tutor — no external API calls, no bills

In-browser adaptive-pacing model powered by TensorFlow.js

Live Teacher/Admin dashboard with class-wide insights

# A 17-Level Journey, Not a Test

*Progressive difficulty keeps students in a flow state — challenged, never overwhelmed*

**Beginner**

### Levels 1–5

Our Environment · Ecology · Natural Resources · Climate & Weather

**Intermediate**

### Levels 6–11

Climate Change · Biodiversity · Pollution · Waste Management · Sustainable Living

**Advanced**

### Levels 12–17

Renewable Energy · Forests & Oceans · Policy & Law · Green Tech

## Progression that keeps students coming back



**Eco Beginner**



**Green Warrior**



**Earth Protector**

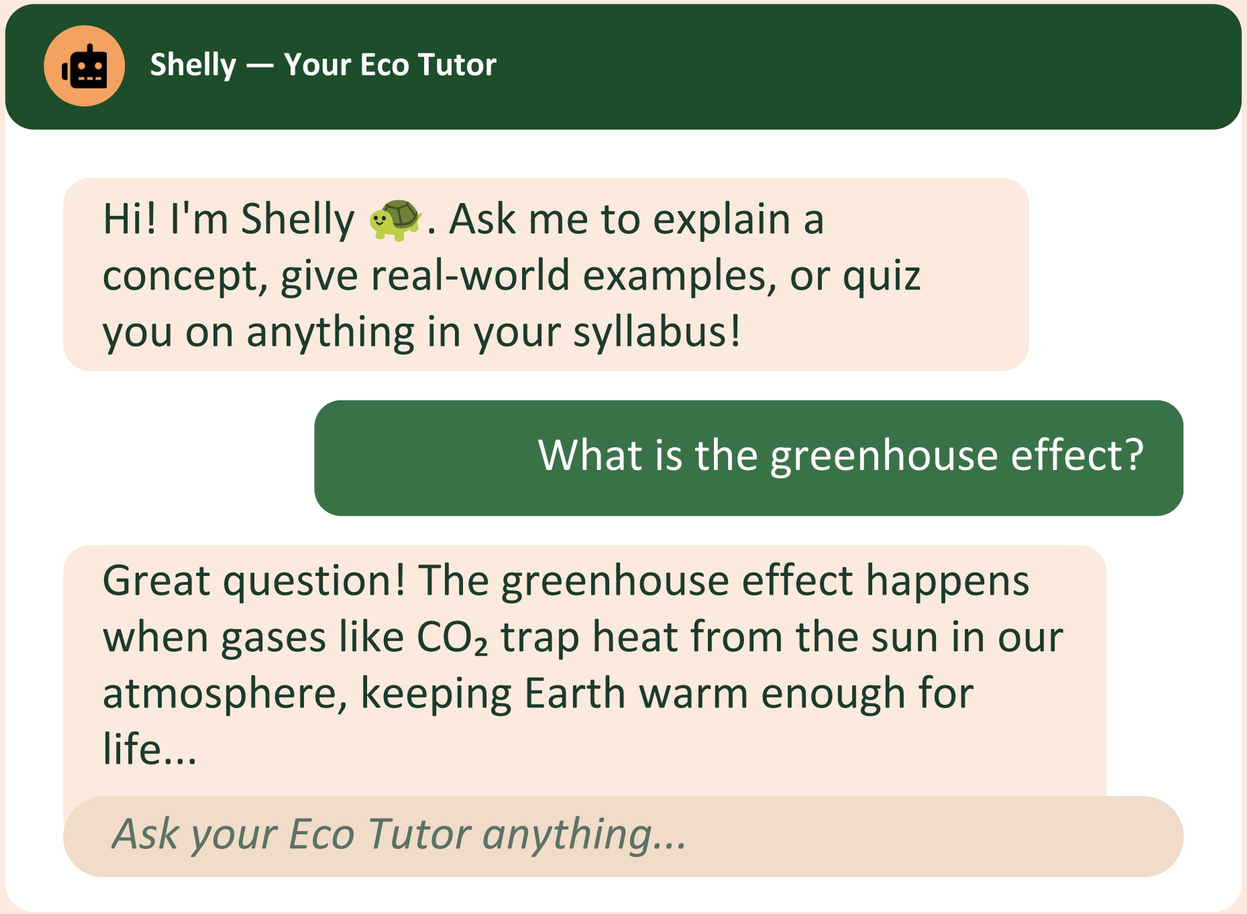


**Eco Champion**

*+ XP, lives, streaks & a real-time Firestore leaderboard*

# Meet Shelly — the Eco Tutor

*A conversational AI tutor that runs 100% locally — answering in seconds, at zero API cost*




Curated knowledge base + keyword matching — no external AI API, no data leaves the browser

Instant answers, even fully offline — no network latency, no downtime

Zero per-query cost — infinitely scalable across every classroom, free forever

# Two More AI-Flavored Modules

*Practical skills and personalized pacing — both computed entirely on-device*

## Eco Waste Identification

Tap a sample waste card — plastic bottle, banana peel, battery — to instantly see its category, recyclability, and disposal tips.

Plastic Bottle | Banana Peel | Battery | +more

*No photo upload. No cloud vision API. No storage cost.*

→ **Logged straight to the class leaderboard via Firestore.**

## Adaptive Learning Engine

A lightweight TensorFlow.js network trains right in the browser on each class's real quiz history from Firestore.

**Accelerate** | **Steady** | **Review**

Suggests the right pacing for every student — automatically, in real time.

*No GPU. No inference bill. Runs on any device with a browser.*

# Built for Teachers, Not Just Students

*A live analytics dashboard turns quiz data into classroom decisions*

### Total Students

Live headcount across every class

### Average Score

Class-wide performance at a glance

### Top Performer

Auto-surfaced from the leaderboard

### At-Risk Students

Rule-based flags for early intervention

- Level-completion & language-usage breakdowns
- Rule-based class-wide “AI Insights” panel
- One-click downloadable CSV reports

# Enterprise-Grade Thinking, Zero Cost

*Runs entirely on Firebase's free Spark plan — no billing account, no credit card, no paid AI API, ever*

| | |
|---|---|
| Firebase Storage — NOT USED | Firestore (free tier) — IN USE |
| Cloud Functions — NOT USED | Firebase Hosting (free tier) — IN USE |
| External AI APIs (GPT / Gemini / Vertex) — NOT USED | TensorFlow.js — 100% in-browser — IN USE |
| Firebase Authentication (billable tier) — NOT USED | |

**The pitch in one line:** every “AI” feature — the tutor, the waste cards, the adaptive pacing, the teacher insights — runs on curated logic and on-device ML, so cost stays at $0 no matter how many schools adopt it.

# Simple, Serverless, Scalable

*A lean architecture any engineering team can audit in an afternoon*



## Frontend

Self-contained HTML/CSS/JS — zero build step, deploys straight from Firebase Hosting



## Data Layer

Firestore — 2 lean collections (scores, waste_logs) power the leaderboard & dashboard



## On-Device AI

TensorFlow.js for adaptive pacing; rule-based NLP for the Eco Tutor & insights



## Security

Locked-down Firestore rules — public create, no update/delete — plus a gated teacher login

# Designed to Scale From Day One

*Because cost stays flat, adoption can grow without a matching budget line*

**Projected Infra Cost**

**$0**

*per month — at any scale*

| $0 | $0 | $0 | $0 |
|---|---|---|---|
| 100 Students | 1,000 Students | 10,000 Students | 100,000 Students |

### Multi-school ready

Same Firestore project scales to any number of classes with no code change

### Bilingual today, more tomorrow

English & Tamil now — the local knowledge base pattern extends to any language

### Data teachers actually use

CSV export + rule-based insights mean adoption doesn't require training

# Already Built. Already Working.

*This isn't a concept deck — it's a functioning product with documented setup, deployment, and security*

### Full working app

index.html student app + teacher.html admin dashboard, both live-tested end to end

### Deployment-ready

Documented INSTALLATION, FIREBASE_SETUP & DEPLOYMENT guides — spin up a new instance in minutes

### Security reviewed

Firestore rules restrict writes to create-only; teacher access is password-gated

### Analytics that ship

Teacher dashboard already computes averages, at-risk flags, and CSV export — not a mockup

# What's Next

*A clear, low-cost path from prototype to district-wide rollout*

1

## Pilot

Deploy in 2–3 partner classrooms; gather real usage & score data

2

## Expand Content

Grow the knowledge base and add more regional languages

3

## Partner

Work with school boards / CSR programs for district-wide rollout

4

## Measure Impact

Publish learning-outcome data to prove behavior change, not just quiz scores



**What we're asking for:** a chance to pilot EcoChronicles with your CSR / education-outreach team — proof of concept is already live, the only missing piece is real classrooms.

# Thank You

*Let's make environmental education something students actually want to open.*

Team EcoChronicles